30/01/2023
Evening

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2023 (Online) Phase-1

(Physics, Chemistry and Mathematics)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) The Test Booklet consists of 90 questions. The maximum marks are 300.

(3) There are **three** parts in the question paper consisting of **Physics, Chemistry** and **Mathematics** having 30 questions in each part of equal weightage. Each part (subject) has two sections.

(i) **Section-A:** This section contains 20 multiple choice questions which have only one correct answer. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

(ii) **Section-B:** This section contains 10 questions. In Section-B, attempt any **five questions out of 10.** The answer to each of the questions is a numerical value. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer. For Section-B, the answer should be rounded off to the nearest integer.

PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. A force is applied to a steel wire 'A', rigidly clamped at one end. As a result elongation in the wire is 0.2 mm. If same force is applied to another steel wire 'B' of double the length and a diameter 2.4 times that of the wire 'A', the elongation in the wire 'B' will be (wires having uniform circular cross-sections)

(1) 6.9×10^{-2} mm (2) 3.0×10^{-2} mm

(3) 6.06×10^{-2} mm (4) 2.77×10^{-2} mm

Answer (1)

Sol. $\because \quad \Delta\ell = \dfrac{F\ell(4)}{Y\pi d^2}$

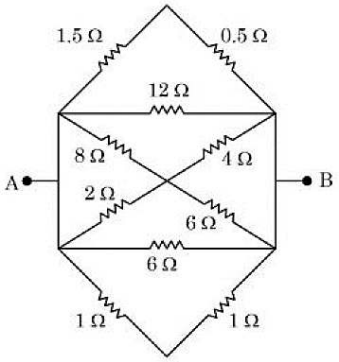
$$\frac{\Delta\ell_1}{\Delta\ell_2} = \frac{\ell_1}{d_1^2} \times \frac{d_2^2}{\ell_2}$$

$$\frac{0.2}{\Delta\ell_2} = \frac{1}{2} \times (2.4)^2$$

$$\Delta\ell_2 = \frac{2 \times 0.2}{(2.4)^2}$$

$= 6.9 \times 10^{-2}$ mm

2. The equivalent resistance between A and B is

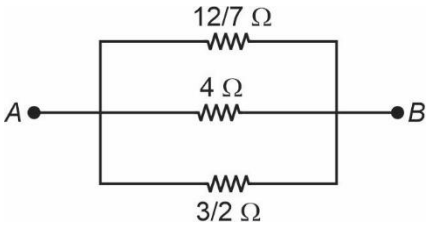


(1) $\frac{2}{3}\Omega$ (2) $\frac{1}{2}\Omega$

(3) $\frac{3}{2}\Omega$ (4) $\frac{1}{3}\Omega$

Answer (1)

Sol. Equivalent circuit can be drawn as

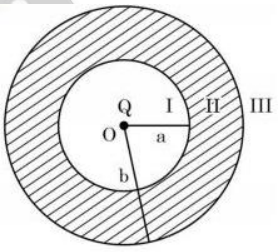


$$\therefore \quad \frac{1}{R_{AB}} = \frac{7}{12} + \frac{1}{4} + \frac{2}{3}$$

$$\boxed{R_{AB} = \frac{2}{3}\Omega}$$

3. As shown in the figure, a point charge Q is placed at the centre of conducting spherical shell of inner radius a and outer radius b. The electric field due to charge Q in three different regions I, II and III is given by:

(I : $r < a$, II : $a < r < b$, III : $r > a$)



(1) $E_I \neq 0, E_{II} = 0, E_{III} \neq 0$

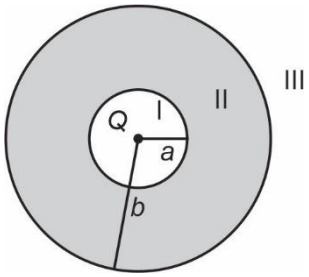
(2) $E_I = 0, E_{II} = 0, E_{III} = 0$

(3) $E_I \neq 0, E_{II} = 0, E_{III} = 0$

(4) $E_I = 0, E_{II} = 0, E_{III} \neq 0$

Answer (1)

Sol.



$E_I \neq 0$ (inside region)

$E_{II} = 0$ (conducting region)

$E_{III} \neq 0$

$= \dfrac{KQ}{r^2} \quad (r > b)$

4. Match List I with List II:

	List I		List II
A.	Torque	I.	kg m^{-1} s^{-2}
B.	Energy density	II.	kg ms^{-1}
C.	Pressure gradient	III.	kg m^{-2} s^{-2}
D.	Impulse	IV.	kg m^2 s^{-2}

Choose the ***correct*** answer from the options given below:

(1) A-IV, B-I, C-II, D-III

(2) A-IV, B-III, C-I, D-II

(3) A-IV, B-I, C-III, D-II

(4) A-I, B-IV, C-III, D-II

Answer (3)

Sol. Torque → kg m^2 s^{-2} (IV)

Energy density → kg m^{-1} s^{-2} (I)

Pressure gradient → kg m^{-2} s^{-2}(III)

Impulse → kg m s^{-1} (II)

5. Match List I with List II:

	List I		List II
A.	Attenuation	I.	Combination of a receiver and transmitter.
B.	Transducer	II.	Process of retrieval of information from the carrier wave at receiver
C.	Demodulation	III.	Converts one form of energy into another
D.	Repeater	IV.	Loss of strength of a signal while propagating through a medium

Choose the ***correct*** answer from the options given below:

(1) A-IV, B-III, C-I, D-II

(2) A-II, B-III, C-IV, D-I

(3) A-IV, B-III, C-II, D-I

(4) A-I, B-II, C-III, D-IV

Answer (3)

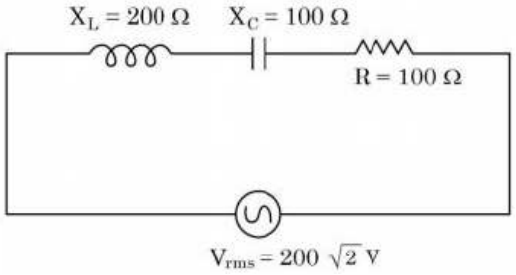
Sol. Theoretical attenuation → (IV)

Transducer → (III)

Demodulation → (II)

Repeater → (I)

6. In the given circuit, rms value of current (I_{rms}) through the resistor R is



(1) 20 A

(2) $2\sqrt{2}$ A

(3) 2 A

(4) $\frac{1}{2}$ A

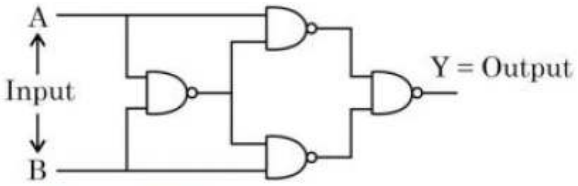
Answer (3)

Sol. $I_{rms} = \frac{V_{rms}}{z} = \frac{200\sqrt{2}}{\sqrt{100^2 + (200-100)^2}}$

$= \frac{200\sqrt{2}}{100\sqrt{2}}$

$= 2$ A

7. The output Y for the inputs A and B of circuit is given by



Truth table of the shown circuit is

(1)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(3)

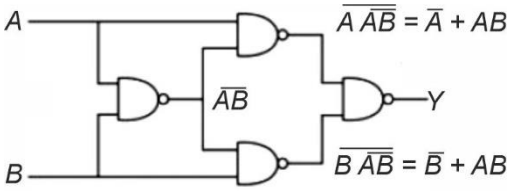
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(4)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Answer (2)

Sol.



$$Y = \overline{\left(\overline{A} + AB\right)\left(\overline{B} + AB\right)} = \left(A + B\right)\left(\overline{A} + \overline{B}\right)$$

$= A\overline{B} + B\overline{A}$ (XOR gate)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

8. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: Efficiency of a reversible heat engine will be highest at –273°C temperature of cold reservoir.

Reason R: The efficiency of Carnot's engine depends not only on temperature of cold reservoir but it depends on the temperature of hot reservoir too and is given as $\eta = \left(1 - \frac{T_2}{T_1}\right)$.

In the light of the above statements, choose the **correct** answer from the options given below

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**

(2) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**

(3) **A** is false but **R** is true

(4) **A** is true but **R** is false

Answer (1)

Sol. $\eta = 1 - \frac{T_{cold}}{T_{hot}}$

T_{cold} = OK $\Rightarrow \eta = 1$ (max)

A is correct.

R is also correct and explains A.

9. A vehicle travels 4 km with speed of 3 km/ h and another 4 km with speed of 5 km/h, then its average speed is

(1) 4.00 km/h (2) 4.25 km/h

(3) 3.50 km/h (4) 3.75 km/h

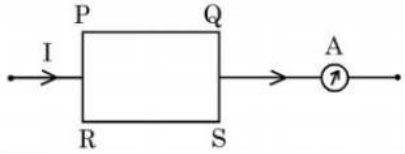
Answer (4)

Sol. Average speed $= \frac{\text{Total distance}}{\text{Total time}}$

$$= \frac{8}{\frac{4}{3} + \frac{4}{5}}$$

= 3.75 km/h

10. A current carrying rectangular loop *PQRS* is made of uniform wire. The length $PR = QS = 5$ cm and $PQ = RS = 100$ cm. If ammeter current reading changes from I to $2I$, the ratio of magnetic forces per unit length on the wire *PQ* due to wire *RS* in the two cases respectively $\left(f_{PQ}^{I} : f_{PQ}^{2I}\right)$ is



(1) 1 : 2 (2) 1 : 4

(3) 1 : 3 (4) 1 : 5

Answer (2)

Sol. Force between two current carrying wire

$$= \frac{\mu_0 I_1 I_2}{2\pi d} \times L$$

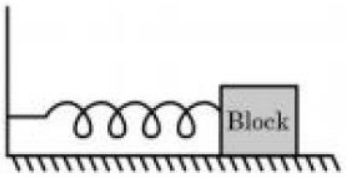
Here I_1 & I_2 are equal

$$F = \frac{\mu_0 I^2}{2\pi d} \times L$$

$F \propto I^2$

$$\frac{F_I}{F_{2I}} = \frac{I^2}{4I^2} = \frac{1}{4}$$

11. For a simple harmonic motion in a mass spring system shown, the surface is frictionless. When the mass of the block is 1 kg, the angular frequency is ω_1. When the mass block is 2 kg, the angular frequency is ω_2. The ratio ω_2/ω_1 is



(1) $\frac{1}{2}$ (2) 2

(3) $\sqrt{2}$ (4) $\frac{1}{\sqrt{2}}$

Answer (4)

Sol. $\omega = \sqrt{\frac{K}{m}} \Rightarrow \omega \propto \frac{1}{\sqrt{m}}$

$$\frac{\omega_2}{\omega_1} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{1}{2}}$$

12. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: The nuclear density of nuclides $^{10}_{5}B$, $^{6}_{3}Li$, $^{56}_{26}Fe$, $^{20}_{10}Ne$ and $^{209}_{83}Bi$ can be arranged as $\rho^N_{Bi} > \rho^N_{Fe} > \rho^N_{Ne} > \rho^N_{B} > \rho^N_{Li}$

Reason R : The radius R of nucleus is related to its mass number A as $R = R_0 A^{1/3}$, where R_0 is a constant.

In the light of the above statements, choose the correct answer from the options given below

(1) Both **A** and **R** are true but **R** is **Not** the correct explanation of **A**

(2) **A** is true but **R** is false

(3) Both **A** and **R** are true and **R** is the correct explanation of **A**

(4) **A** is false but **R** is true

Answer (4)

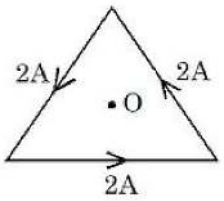
Sol. $R = R_0 A^{\frac{1}{3}}$, using this

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} = \frac{Am_P}{\frac{4}{3}\pi R_0^3 A} = \frac{m_P}{\frac{4}{3}\pi R_0^3}$$

ρ is independent of mass number.

$\therefore$ A is false

13. As shown in the figure, a current of 2 A flowing in an equilateral triangle of side $4\sqrt{3}$ cm. The magnetic field at the centroid O of the triangle is



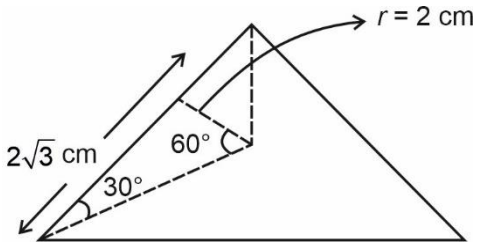
(Neglect the effect of earth's magnetic field)

(1) $3\sqrt{3}\times10^{-5}$ T (2) $\sqrt{3}\times10^{-4}$ T

(3) $4\sqrt{3}\times10^{-5}$ T (4) $4\sqrt{3}\times10^{-4}$ T

Answer (1)

Sol.



$$B_{net} = \frac{\mu_0 i}{4\pi r}(\sin\alpha + \sin\beta)\times 3$$

$$= \frac{\mu_0 \times 2}{4\pi\times\left(2\times10^{-2}\right)}\times\left(\frac{\sqrt{3}}{2}+\frac{\sqrt{3}}{2}\right)\times 3$$

$$= 10^{-7}\times10^{2}\left(3\sqrt{3}\right)$$

$$= 3\sqrt{3}\times10^{-5}\,\text{T}$$

14. An object is allowed to fall from a height R above the earth, where R is the radius of earth. Its velocity when it strikes the earth's surface, ignoring air resistance, will be

(1) $\sqrt{\frac{gR}{2}}$

(2) $2\sqrt{gR}$

(3) $\sqrt{gR}$

(4) $\sqrt{2gR}$

Answer (3)

Sol. $U_P = -\frac{GMm}{2R}$

$$U_S = -\frac{GMm}{R}$$

$\Rightarrow$ Energy conservation

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = -\frac{GMm}{2R}$$

$$v^2 = \frac{GM}{R}$$

$$v = \sqrt{\frac{GM}{R}} = \sqrt{gR}$$

15. A flask contains hydrogen and oxygen in the ratio of 2 : 1 by mass at temperature 27°C. The ratio of average kinetic energy per molecule of hydrogen and oxygen respectively is:

(1) 1 : 1 (2) 2 : 1

(3) 4 : 1 (4) 1 : 4

Answer (1)

Sol. K.E. per molecule = $\left(\frac{f}{2}KT\right)$

$$\frac{\text{average}(K.E)_{\text{hydrogen}}}{\text{average}(K.E)_{\text{oxygen}}} = \frac{f_{\text{hydrogen}}}{f_{\text{oxygen}}} = 1$$

16. A thin prism P_1 with an angle 6° and made of glass of refractive index 1.54 is combined with another prism P_2 made from glass of refractive index 1.72 to produce dispersion without average deviation. The angle of prism P_2 is

(1) 6°

(2) 1.3°

(3) 4.5°

(4) 7.8°

Answer (3)

Sol. $(\mu_1 - 1)A_1 = (\mu_2 - 1)A_2$

$\Rightarrow (1.54 - 1)6 = (1.72 - 1)A_2$

$$A_2 = \left(\frac{0.54}{0.72} \times 6\right) = \frac{18}{4} = \left(\frac{9}{2}\right) = 4.5°$$

17. A point source of 100 W emits light with 5% efficiency. At a distance of 5 m from the source, the intensity produced by the electric field component is:

(1) $\frac{1}{40\pi}\frac{W}{m^2}$

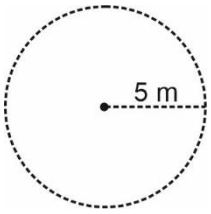
(2) $\frac{1}{2\pi}\frac{W}{m^2}$

(3) $\frac{1}{20\pi}\frac{W}{m^2}$

(4) $\frac{1}{10\pi}\frac{W}{m^2}$

Answer (1)

Sol.



Intensity at 5 m = $\frac{5}{4\pi \times 5^2}\left(\frac{W}{m^2}\right)$

$$= \frac{1}{20\pi}\left(\frac{W}{m^2}\right)$$

Intensity due to electric field $= \frac{1}{40\pi}\left(\frac{W}{m^2}\right)$

$$= \left(\frac{W}{40\pi}\right)$$

18. An electron accelerated through a potential difference V_1 has a de-Broglie wavelength of λ. When the potential is changed to V_2. Its de-Broglie wavelength increases by 50%. The value of $\left(\frac{V_1}{V_2}\right)$ is equal to

(1) $\frac{3}{2}$

(2) $\frac{9}{4}$

(3) 4

(4) 3

Answer (2)

Sol. $P = \sqrt{2\,eVm}$

$$\lambda = \left(\frac{h}{P_1}\right) \quad \ldots(i)$$

$$\frac{3\lambda}{2} = \frac{h}{P_2} \quad \ldots(ii)$$

Dividing (i) by (ii)

$$\Rightarrow \frac{2}{3} = \left(\frac{P_2}{P_1}\right) = \sqrt{\frac{v_2}{v_1}}$$

$$\Rightarrow \frac{4}{9} = \left(\frac{v_2}{v_1}\right)$$

$$\frac{v_1}{v_2} = \left(\frac{9}{4}\right)$$

19. A machine gun of mass 10 kg fires 20 g bullets at the rate of 180 bullets per minute with a speed of 100 m s^{-1} each. The recoil velocity of the gun is

(1) 0.6 m/s

(2) 0.02 m/s

(3) 1.5 m/s

(4) 2.5 m/s

Answer (1)

Sol. Momentum of bullets per unit time

$$= \frac{180 \times \frac{20}{1000} \times 100}{60} \text{ kg m/s}^2$$

$$= 6 \text{ N}$$

$\Rightarrow$ Force on gun = 6 N

We cannot calculate recoil velocity with the given data.

If we consider recoil velocity at $t = 1$ s, then

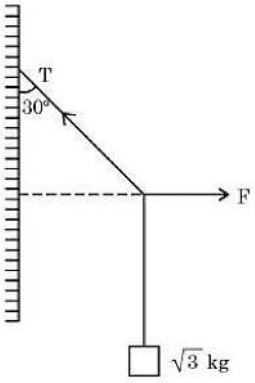
$V_{recoil} = u + at$

$$= 0 + \frac{6}{10} \times 1$$

$$= 0.6 \text{ m/s}$$

20. A block of $\sqrt{3}$ kg is attached to a string whose other end is attached to the wall. An unknown force F is applied so that the string makes an angle of 30° with the wall. The tension T is:

(Given g = 10 ms^{-2})



(1) 15 N

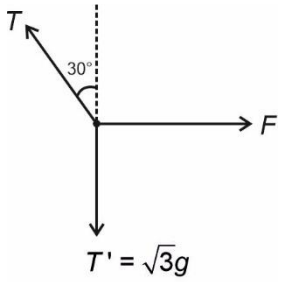
(2) 20 N

(3) 10 N

(4) 25 N

Answer (2)

Sol. Drawing the FBD of the point where *F* is applied



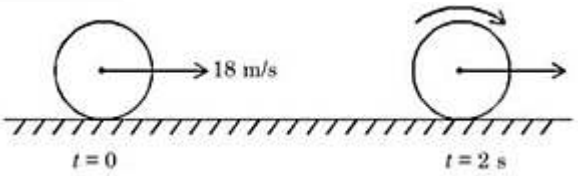
$\Rightarrow T\cos 30° = \sqrt{3}\, g$

$\Rightarrow T = 2g = 20$ N

SECTION - B

Numerical Value Type Questions: This section contains 10 questions. In Section B, attempt any five questions out of 10. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30) using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.

21. A uniform disc of mass 0.5 kg and radius *r* is projected with velocity 18 m/s at $t = 0$ s on a rough horizontal surface. It starts off with a purely sliding motion kinetic energy of the disc after 2 s will be_____ J (given, coefficient of friction is 0.3 and $g = 10$ m/s^2).



Answer (54)

Sol. $v = v_0 - \mu g t$

$\Rightarrow v = 18 - 0.3 \times 10 \times 2 = 12$ m/s

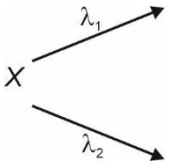
$\Rightarrow$ Kinetic energy $= \frac{1}{2}mv^2 + \frac{1}{2}\frac{mv^2}{2}$

$$= \frac{3}{4}mv^2 = \frac{3}{4} \times 0.5 \times 144 \text{ J} = 54 \text{ J}$$

22. A radioactive nucleus decays by two different process. The half life of the first process is 5 minutes and that of the second process is 30 s. The effective half-life of the nucleus is calculated to be $\frac{a}{11}$ s. The value of *a* is _____.

Answer (300)

Sol.



$\Rightarrow \lambda_{eff} = \lambda_1 + \lambda_2$

$$\Rightarrow \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{(t_{1/2})_1} + \frac{\ln 2}{(t_{1/2})_2}$$

$\Rightarrow \quad t_{1/2} = \dfrac{(t_{1/2})_1 \times (t_{1/2})_2}{(t_{1/2})_1 + (t_{1/2})_2} = \dfrac{300 \times 30}{300 + 30}\text{s} = \dfrac{300}{11}\text{s}$

$\Rightarrow \quad \alpha = 300$

23. A faulty thermometer reads 5°C in melting ice and 95°C in stream. The correct temperature on absolute scale will be _____ K when the faulty thermometer reads 41°C.

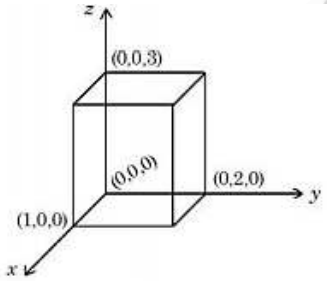
Answer (313)

Sol. let the correct temperature be X°C

$\Rightarrow \quad \dfrac{X-0}{100-0} = \dfrac{41-5}{95-5} \Rightarrow X = 40$

$\Rightarrow$ Temperature is 273 + 40 K = 313 K

24. As shown in figure, a cuboid lies in a region with electric field $\vec{E} = 2x^2\hat{i} - 4y\hat{j} + 6\hat{k}\ \dfrac{N}{C}$. The magnitude of charge within the cuboid is $n\in_0 C$. The value of n is _____ (if dimension of cuboid is 1 × 2 × 3 m^3).



Answer (12)

Sol. Flux through planes parallel to y-z = $2(1)^2$ × Area

= $2(1)^2 \times 2 \times 3$

= 12 Nm^2/C

Flux through planes parallel to x-z = –4(2) × Area

= –4(2) × 1 × 3

= –24 Nm^2/C

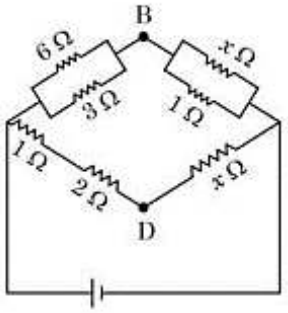
Flux through planes parallel to x-y = 0

$\Rightarrow \quad \phi_{Total} = 12 - 24 = -12$

$\Rightarrow \quad -12 = \dfrac{q_{enc}}{\varepsilon_0} \Rightarrow |q_{enc}| = 12\varepsilon_0$

$\Rightarrow \quad n = 12$

25. If the potential difference between B and D is zero, the value of x is $\dfrac{1}{n}\Omega$. The value of n is _____.



Answer (2)

Sol. The circuit is a wheat stone bridge, so

$$\frac{\dfrac{6\times 3}{6+3}}{\dfrac{x\times 1}{x+1}} = \frac{1+2}{x}$$

$\Rightarrow \quad \dfrac{2(x+1)}{x} = \dfrac{3}{x}$

$\Rightarrow \quad x = \dfrac{1}{2}$

So $n = 2$

26. The velocity of a particle executing SHM varies with displacement (x) as $4v^2 = 50 - x^2$. The time period of oscillation is $\dfrac{x}{7}$s. The value of x is _____.

$\left(\text{Take } \pi = \dfrac{22}{7}\right)$

Answer (88)

Sol. $4v^2 = 50 - x^2$

or $v = \dfrac{1}{2}\sqrt{50 - x^2}$

Comparing the above equation with $v = \omega\sqrt{A^2 - x^2}$

$\Rightarrow \omega = \dfrac{1}{2}$ & $A = \sqrt{50}$

so $\dfrac{2\pi}{T} = \dfrac{1}{2}$

$\Rightarrow T = 4\pi$ sec

$= 4 \times \dfrac{22}{7}$sec

$T = \dfrac{88}{7}$sec

so $x = 88$

27. In an ac generator, a rectangular coil of 100 turns each having area 14 × 10^{-2} m^2 is rotated at 360 rev/min about an axis perpendicular to a uniform magnetic field of magnitude 3.0 T. The maximum value of the emf produced will be _____ V. $\left(\text{Take } \pi = \frac{22}{7}\right)$

Answer (1584)

Sol. $\phi = B.A$

$\phi = BNA\cos\omega t$

So $Emf = \frac{-d\phi}{dt} = NBA\omega \sin\omega t$

So maximum value of emf is

$E_{max} = NBA\omega$

$= 100 \times 3 \times 14 \times 10^{-2} \times \frac{360 \times 2\pi}{60}$

= 1584

28. A body of mass 2 kg is initially at rest. It starts moving unidirectionally under the influence of a source of constant power *P*. Its displacement in 4 s is $\frac{1}{3}\alpha^2\sqrt{P}$ m. The value of α will be _____.

Answer (04)

Sol. $P = Fv$

$m\frac{vdv}{dt} = P$

$m\int_0^v vdv = \int_0^t Pdt$

$\frac{mv^2}{2} = Pt$

$v = \sqrt{\frac{2P}{m}}t^{1/2}$

$\int_0^s ds = \sqrt{\frac{2P}{m}}\int_0^t t^{1/2}dt$

$s = \frac{2}{3}\sqrt{\frac{2P}{m}}t^{3/2}$

or $s = \frac{2}{3}\sqrt{\frac{2P}{2}} \times 4^{3/2}$

$= \frac{16}{3}\sqrt{P}\,m$

So $\alpha = 4$

29. In a Young's double slit experiment, the intensities at two points, for the path differences $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$ [λ being the wavelength of light used) are I_1 and I_2 respectively. If I_0 denotes the intensity produced by each one of the individual slits, then $\frac{I_1 + I_2}{I_0}$ = _____.

Answer (03)

Sol. $I' = I\cos^2\left(\frac{k\Delta x}{2}\right)$

so $I_1 = 4I_0\cos^2\left(\frac{2\pi}{2\lambda} \times \frac{\lambda}{4}\right)$

$I_1 = 2I_0$

& $I_2 = 4I_0\cos^2\left(\frac{2\pi}{2\lambda} \times \frac{\lambda}{3}\right)$

$I_2 = I_0$

So $\frac{I_1 + I_2}{I_0} = 3$

30. A stone tied to 180 cm long string at its end is making 28 revolutions in horizontal circle in every minute. The magnitude of acceleration of stone is $\frac{1936}{x}ms^{-2}$. The value of *x* _____. $\left(\text{Take } \pi = \frac{22}{7}\right)$

Answer (125)

Sol. Acceleration of stone $a = \frac{v^2}{r} = \omega^2 R$

$a = \left(\frac{28 \times 2}{60} \times \frac{22}{7}\right)^2 \times 1.8$

$= \frac{1936}{125}$

So $x = 125$

CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

31. Match **List I** with **List II**:

	List I (Complexes)		**List II (Hybridisation)**
A.	$[Ni(CO)_4]$	I.	sp^3
B.	$[Cu(NH_3)_4]^{2+}$	II.	dsp^2
C.	$[Fe(NH_3)_6]^{2+}$	III.	sp^3d^2
D.	$[Fe(H_2O)_6]^{2+}$	IV.	d^2sp^3

(1) A-II, B-I, C-IV, D-III (2) A-I, B-II, C-IV, D-III

(3) A-I, B-II, C-III, D-IV (4) A-II, B-I, C-III, D-IV

Answer (2)

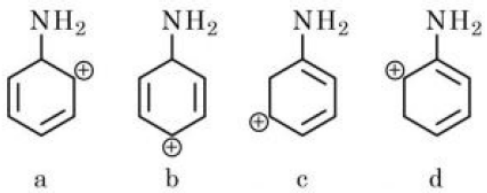
Sol. $Ni(CO)_4 \longrightarrow sp^3$

$[Cu(NH_3)_4]^{2+} \longrightarrow dsp^2$

$[Fe(NH_3)_6]^{2+} \longrightarrow d^2sp^3$

$[Fe(H_2O)_6]^{2+} \longrightarrow sp^3d^2$

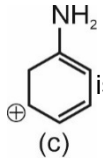
32. The most stable carbocation for the following is:



(1) a (2) b

(3) d (4) c

Answer (3)

Sol.  is the most stable carbocation because its resonance goes up to nitrogen atom.

(c)

33. Which of the following reaction is correct?

(1) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 2N_2O_4 + O_2$

(2) $2LiNO_3 \longrightarrow 2Li + 2NO_2 + O_2$

(3) $2LiNO_3 \xrightarrow{\Delta} 2NaNO_2 + O_2$

(4) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2 + O_2$

Answer (4)

Sol. $4LiNO_3 \xrightarrow{Heat} 2Li_2O + 4NO_2 + O_2$

$LiNO_3$ on heating produces Li_2O, NO_2 and O_2

34. The wave function (Ψ) of 2s is given by

$$\Psi_{2s} = \frac{1}{2\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{1/2}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0}$$

At r = r_0, radial node is formed. Thus, r_0 in terms of a_0

(1) $r_0 = 2a_0$ (2) $r_0 = 4a_0$

(3) $r_0 = a_0$ (4) $r_0 = \frac{a_0}{2}$

Answer (1)

Sol. For radial node $\Psi_{2s} = 0$

$\therefore r = 2a_0$

35. 1 L, 0.02 M solution of $[Co(NH_3)_5SO_4]$ Br is mixed with 1 L, 0.02 M solution of $[Co(NH_3)_5Br]SO_4$. The resulting solution is divided into two equal parts (X) and treated with excess of $AgNO_3$ solution and $BaCl_2$ solution respectively as shown below:

1 L Solution (X) + $AgNO_3$ solution (excess) $\rightarrow$ Y

1 L Solution (X) + $BaCl_2$ solution (excess) $\rightarrow$ Z

The number of moles of Y and Z respectively are

(1) 0.02, 0.02 (2) 0.01, 0.01

(3) 0.02, 0.01 (4) 0.01, 0.02

Answer (2)